Dental Replacement in Gorgonopsia (Therapsida) Studied Using X-ray Microtomography: Implications for the Ancestral Theriodont Dental Replacement Pattern

Introduction

Gorgonopsia is an extinct clade of carnivorous, sabre-toothed therapsids. They are distinguished by their vomerine ridges and enlarged canine and incisor teeth. According to current theories, gorgonopsians were fully terrestrial and had a semi-erect gait, comparable to that of modern crocodilians (Benoit, et al., 2023). It is thought that gorgonopsians fed on giant pareiasaurs, which are heavy-armoured creatures that resemble turtles, and dinocephalians, and other species of synapsids (Kemp, 2005). Gorgonopsians were probably active apex predators. There are six major clades of therapsids, namely Biarmosuchia, Anomodontia, Dinocephalia, Gorgonopsia, Therocephalia, and Cynodontia (Rubidge and Sidor, 2001; Kammerer et al., 2014; Angielczyk and Kammerer 2018; Kammerer, et al., 2022). Gorgonopsians are believed to be related to Cynodontia and Therocephalia (the former including modern days mammals) in a clade called Theriodontia (Kammerer et al., 2014). It is documented that the extinction of Gorgonopsia was caused by volcanic eruptions that formed the Siberian traps during the End-Permian mass extinction event, leading to the emission of greenhouse gases, increased temperatures, acid rains, and depletion of the ozone layer leading to more wildfires (Benton, 1997).

The first known gorgonopsian remains from the Beaufort Group of the Karoo Supergroup of South Africa were described by palaeontologist Richard Owen in 1876. The specimen was named Gorgonops torvus and became the namesake of the order Gorgonopsia (Owen, 1876). Gorgonopsians were subsequently discovered in Russia (Kammerer and Masyutin, 2018), China(Young, 1979), and other countries in Africa (e.g. Tanzania, Niger, Zambia, and Malawi), thus demonstrating a near global distribution. Gorgonopsians are classified as synapsids one of the two major clade of Amniote which are diagnosed by the presence of a single temporal fenestra, i.e., an opening in the skull roof located behind the orbit (Owen, 1876). The fenestra is a significant anatomical structure that gives the jaw muscles more room inside the skull for attachment (Levy, 2023). Synapsids include living and extinct mammal species, as well as the “mammal-like reptiles”, i.e. Therapsida. Non-mammalian therapsids lived mostly between 270 and 200 million years ago, during the Permian and Triassic periods and are the ancestors of mammals (Kemp, 2006). The development of the mammalian auditory apparatus, sequences of tooth replacement, breathing, endothermic metabolism, locomotion, and the formation of the mammalian brain are just a few of the traits that connect therapsid to mammals (Laaß and Schillinger, 2015; Norton et al., 2023).

Gorgonopsia are consensually considered the sister taxon to the Cynodontia and Therocephalia in a clade called the Theriodontia. They are themselves divided into three subfamilies that include the Gorgonopsinae, Rubidgeinae, and Inostranceviinae, along with a number of specimens that are currently not assigned to a higher clade due to uncertain taxonomic affinities (Kieser et al., 1993; Kammerer, 2016). The first published phylogeny of selected species was produced by Kammerer (2016) and updated regularly since then (e.g., Kammerer and Rubidge, 2022). Gorgonopsia are most famous for including the largest carnivores of the late Permian, such as Inostrancevia and Rubidgea, that were bear-sized and sabre-toothed, with a skull length up to 60 cm (Sigogneau-Russell, 1989; Kammerer et al., 2014) whereas others were as small as a domestic cat.

One of the oldest gorgonopsian taxa is Eriphostoma microdon, first named in 1911 by palaeontologist Robert Broom (Broom,1911). In 2014, after CT scanning the holotype, Eriphostoma microdon was revised and accepted as a valid genus and species of gorgonopsian, making it one of the best defined taxon among gorgonopsians (Kammerer et al., 2014, Kammerer, 2015). Eriphostoma is also among the smallest known gorgonopsians, with a skull length of less than 15 cm. Like many other gorgonopsians, it has five incisors, one upper and lower canine and three

small post canine teeth. The diagnostic features of Eriphostoma microdon include the very small size of its incisors compared to other gorgonopsians, its relatively short and wide snout, the presence of a shallow supraorbital depression, and the narrow space between its post-canine teeth, which appear tightly packed in lateral view (Kammerer et al., 2014, Kammerer, 2015). Eriphostoma being a very small gorgonopsian and the only one found so far in the upper Tapinocephalus Assemblage Zone (AZ), it marks a peculiar phase in gorgonopsian evolution at a time when they were limited to small-bodied animals, whereas earlier in the Eodicynodon AZ and later in the late Permian (Cistecephalus and Daptocephalus AZs) they would attain very large body sizes (Rubidge and Kammerer, 2022).

Studying tooth replacement in therapsids is crucial for understanding the evolutionary history and functional adaptations of these ancient vertebrates. Therapsids offer significant insights into the transition from reptilian to mammalian dental structures. By examining their tooth replacement patterns, researchers can trace the evolution of more complex dental systems, such as the development of distinct tooth types like incisors, canines, and molars (LeBlanc et al., 2018; Norton et al 2023). Analysing tooth replacement provides valuable information about the functional adaptations of therapsids. It sheds light on their dietary habits and how their chewing mechanisms evolved over time (Donoghue et al., 2016). Furthermore, studying these patterns helps in understanding the growth and development of therapsids, offering insights into their life history, growth rates, and stages of development (Godefroit, 1997). Gorgonopsians are pivotal for understanding tooth replacement in therapsids due to their unique evolutionary position and dental characteristics. Gorgonopsians mark a transitional phase between early synapsid reptiles and more advanced therapsids (Whitney et al., 2020). Their dental morphology reveals important changes in tooth replacement mechanisms as they evolved from simpler to more complex dentitions, providing crucial insights into this evolutionary process (Kermack, 1956). Their differentiated teeth, including prominent canines, highlight significant shifts in tooth function and replacement strategies. Their well-preserved fossils provide valuable data on dental development and replacement, contributing to the reconstruction of the evolutionary history of dental features. Overall, gorgonopsians offer key transitional evidence that enhances our understanding of the development of advanced dental systems in therapsids and mammals.

Teeth are one of the most important indicators of an animal’s diet and due to their enamel coating they are one of the best-preserved structures in the fossil record. The late Permian was an age of significant transition in terrestrial vertebrate groups, with therapsids and parareptiles replacing the primitive pelycosaur and amphibian grades that dominated communities in the early and middle Permian (Olson, 1962; Brocklehurst et al., 2017). One of the main distinctions between therapsids and pelycosaur-grade synapsids is the emergence of a heterodont dentition, which means that the set of teeth on the maxilla and dentary bones are not uniform in size and shape. Simpson (1936, p.793) proposed that Permian therapsids (such as the gorgonopsians) could be classified as having "incipient heterodonty" meaning the beginning of heterodonty. Gorgonopsians have a set of identically shaped upper and lower incisors that functioned with a powerful bite to cut through tissue to the bone (Kemp, 1982). A single enlarged canine, located distal to the incisors, is succeeded by the postcanine teeth, usually 4 but up to 8 of them, that are smaller, often canted, and recurved distally. The alternative to the heterodont condition is homodonty, which refers to having similar shaped and sized teeth across the jaws.

Tooth replacement refers to the succession of teeth throughout the lifetime of an individual. Alternate replacement, which consists of teeth being replaced at every second or third position, dates from the Late Carboniferous to early Permian (~310–295 Ma), and is the oldest of the main forms of dental replacement. The skull of the extinct temnospondyl amphibian Eryops was studied by Sawin (1941) who found empty alveoli between functioning teeth, supporting alternate tooth replacement. Bolk (1922) believed that the teeth sit alternatively on the base of the lamina and on the buccal side an exostichos being the latter teeth. Each set would give rise to functional teeth (i.e. a tooth that has a matching element to bite against) (Parrington, 1936).Teeth are replaced as alternate pairs on the jaws from pelycosaurs to cynodonts, were the former tooth is called ‘ endostichos * and the latter ‘exostichos’ replacement seen in most early synapsids. On top of an alternate replacement, therapsids had multiple generations of replacement teeth that were replaced throughout the lifetime (polyphyodonty).

Table 1. This table captures the main types of tooth replacement patterns, examples, and their opposites.

In contrast, mammals have a restricted and definitive replacement of teeth (Norton, 2020). All but the molars and last premolar are replaced in most placental mammals, and only the third deciduous premolar is replaced in marsupials. In most representatives of these groups, replacement occurs at each locus only once. The recurring sequence of teeth that succeeds each other in each alveolus of a common amniote ancestor was abbreviated to form the deciduous and permanent teeth of mammals (Kermack, 1956). This condition is referred to as diphyodonty. The term Zahnreihen was initially used by Woerdeman (1921) to characterise the pattern in which teeth emerge and replace one another. A Zahnreihe is a collection of neighboring teeth from various tooth families and are at a comparable developmental stage. Within single-tooth families, the replacement tooth line contrasts with the diagonal pattern of the Zahnreihe. In this manner, if multiple teeth were exfoliated simultaneously, some of the nearly completed teeth would be prepared to erupt and fill the gap in the tooth row (Woerdeman, 1921).

There have been very few studies that have looked at tooth replacement in Gorgonopsia, with only one known study that focused on tooth replacement in this group undertaken in 1956 by palaeontologist Kermack. In his study he found that tooth replacement occurs lingual with the replacement tooth erupting from the alveolus of the functional tooth and that teeth follow an alternate tooth replacement pattern (Kermarck, 1956). Kermack (1956) emphasised that regarding tooth replacement in theriodonts, there is a lot that is still unknown, and that the main problem was fossil preparation. In past studies, there has been a reliance on destructive sampling techniques, such as serial sectioning and grinding (Hopson, 1964), and therefore specimens suitable for study were restricted to incomplete and poorly preserved material. Advances in preparation techniques have allowed palaeontologists to use weak acids to breakdown rock matrix (Rixon 1949; Toombs 1948). Recent advances in X-ray microcomputed tomography (µCT) have allowed for a wider range of specimens to be included in studies on internal anatomy. As such, tooth replacement has been successfully studied using µCT in several therapsid groups, including Dicynodontia (Olroyd et al., 2021), Therocephalia (Benoit et al., 2017; Norton 2020), and Cynodontia (Abdala et al., 2013; Norton et al., 2020, 2021). Bendel et al. (2018) redescribed a specimen of the gorgonopsian Cynariops using µCT scanning, making it possible to view and describe the replacement teeth. To conclude, besides Kermack (1956) and a few isolated descriptions, the dental replacement pattern of gorgonopsians is mostly unknown.

When comparing primitive and more derived synapsids, basal pelycosaur-grade synapsids usually have two upper canines, whereas most therapsids are described as having a single upper canine (Bendel et al., 2018). The single canine condition is encountered in many therapsids but has, in fact, many exceptions such as Raranimus (Duhamel et al., 2021) and Lycosuchus (Abdala et al., 2013; Norton, 2020). Then, the problem comes with gorgonopsians, in which only the most derived taxa, the rubidgeines, seem to possess double upper canines instead of the most primitive ones as is expected given that this is meant to be the plesiomorphic condition. Hence the study of one of the oldest gorgonopsian, Eriphostoma microdon, from the middle Permian Tapinocephalus AZ, and other juveniles and adults gorgonospians from the late Permian may shed some light on the origin of double canines in therapsids, and the polarity of this character among theriodonts.Here I propose to use µCT scanning on a selection of gorgonopsian skulls from different ages, including Eriphostoma, to reconstruct the gorgonopsian condition of dental replacement and address whether the double canine condition indeed occurs in derived taxa only.

Top of Form

Hypotheses:

Bottom of Form

The hypothesis for this study proposes that Gorgonopsia primitively possess two pairs of upper (maxillary) canines and that juvenile individuals exhibit a higher number of replacement teeth compared to adults. Additionally, it is suggested that there will be no evidence supporting finite canine replacement in Gorgonopsia and that alternating replacement patterns will be observed in the incisors and postcanines among the specimens studied.

Aims:

The aims of this investigation are to explore the ancestral tooth replacement patterns and to determine if the double canine condition is present in primitive Gorgonopsia.

Materials and methods

Specimens

This project includes six specimens of gorgonopsian that lived in the middle Permian (272~ 259.9 ma) and the late Permian (258~252.2 ma).

BP/1/155 an indeterminate gorgonopsian found in 1946 by James Kitching on farm Hoeksplaas in the Western Cape. It includes a medium sized skull from the Cistecephalus AZ.

BP/1/575 is a gorgonopsian identified as Aelurosaurus, which was found in 1947 by James Kitching on farm Leeufontein, Western Cape. It includes a skull from the Cistecephalus AZ.

BP/1/216 is a gorgonopsian identified as Arctops kitchingi, which was found in 1946 by BJ Kitching on farm Ringsfontein, Western Cape. It includes a juvenile skull from the Cistecephalus AZ. This is the smallest gorgonopsian of this sample.

BP/1/709 is the holotype of Aloposaurus, which was found in 1947 by BJ Kitching on farm Swaelkrans, Western Cape. It includes a small skull from the Cistecephalus AZ

BP/1/673 is a Cynarioides gracilis which was found in 1947 by SC Kitching in Vleiplaas, Western Cape. It includes a small skull from the Cistecephalus AZ

AM 3752 is the holotype skull of Broomisaurus now synonymised with Eriphostoma microdon, which was found in 1920 at Mynhartskraal, Western Cape. It is from the Tapinocephalus AZ.

1.2 Morphological measurements

To help assess the stage of maturity of specimens, in addition to cranial fusion and the intensity of dental replacement, five measurements were taken (Figure 1). The maximum cranial length is a straight-line distance from the anteriorly projecting midpoint of the premaxilla to the most posteriorly distant posterior point of the occipital condyle (A). The maximum cranial breath is the maximum width of the skull collected from the most protruding point of the temporal region (B). On the upper jaw the following measurement will be collected: the maxilla bicanine breadth, which is the maximum breadth across the alveolar borders of the maxillary canines (C). The length of the dental series which is the distance between the midpoint of the medial incisors to the end of the last postcanine (D). The antorbital snout length measured from midpoint of the interorbital space to the midpoint of the medial incisors (E), the measurements will be taken using a plastic calliper. The number of postcanines were also counted. For specimens that could not be measured physically, digital measurements were taken using Inkscape software. According to the maximum cranial length the specimens have be grouped into their respective ontogeny stages as follows: early juvenile (less than 100 mm), juvenile (100-119 mm), sub-adult (120-149 mm), adult (150-200 mm), late adult (greater than 200 mm) (Bendel, et al., 2018).

1.3 CT scans

The specimens used in this research were µCT scanned with the Nikon Metrology XTH 225/320 LC dual source industrial CT system at the University of the Witwatersrand Microfocus X-ray CT Facility of the Evolutionary Studies Institute.

Table 2. List of specimens included in this study, their identification and voxel size of the CT scan.

Figure 1: Measurements; A: maximum cranial length, B maximum cranial width, C: maxillary bicanine breadth, D: dental series length and E: antorbital snout length. Modified from Gebauer, 2007: figure 25.

Segmentation

The CT data has been segmented using the dedicated workstations available in the Virtual Imaging in Palaeontology (VIP) Laboratory based at the ESI, Wits. Software packages available in the VIP

Lab include VGStudio MAX 3.2.4 64 bit manufactured by Volume Graphic, was used for 3-D reconstructions.

Using VG Studio

Using VGStudio for segmenting teeth from a CT scan of a skull involved a detailed process to accurately isolate and analyse the dental structures. The process began with obtaining CT scan data in a format compatible with VGStudio, typically TIFF stacks; however, some data were DICOM files. After launching VGStudio, a new project was created by selecting File > New Project, naming the project, and choosing a suitable directory to save it. The CT scan data was then imported by navigating to File > Import, selecting the DICOM files, and allowing VGStudio to process and load the data, for segmentation, the appropriate tool was chosen from the Segmentation module, such as Polygon lasso or Region Growing. The thresholding for region growing was applied to be 10 or less dependent on which parameters gave the most accurate segmentation. Manual editing was performed using tools such as the Brush or Lasso to refine the segmentation and remove artifacts with the Eraser tool. The segmentation was reviewed in various views to ensure accuracy, and any necessary adjustments to parameters were made. Once satisfied, the segmentation was saved and exported in a suitable format such as STL or OBJ for further analysis. For specimens that were not segmented, a transverse slice of both the maxillary and mandible teeth are taken and analysed, teeth with low contrast were highlighted in black to emphasis the shape. Top of FormBottom of Form

Results

Measurement

Table 3. A table providing the skull length in millimetres and giving the estimated age such as early juvenile, juvenile, sub-adult, adult, late adult.

Description of segmentation

From the specimens examined a total of 3 different arrangements of the upper canines were observed. Type 1 is one functional canine (mesial), no replacement and a remnant root distally associated with the canine. Type 2 is one functional canine (distal), one replacement (mesial) and remnant root mesial associated with the functional canine. The last type of arrangement, type 3 is one functional canine (distal), multiple replacements (mesial & distal). The specimens are grouped according to which of the three types of arrangement of the upper canine replacement activity is observed.

Type 1. One functional canine (mesial), no replacement

BP/1/155 (indeterminate gorgonopsian)

Description: The specimen is a complete skull with the mandible preserved in tight occlusion (Fig. 2), most of the teeth are preserved in the jaw, the skull shape is mostly preserved in its original shape with only a slight compression on the right temporal region of the skull.

Incisors – the specimen has a total of five upper incisors preserved in the right premaxilla (I1–I5) and three incisors (I1, I2, I3) preserved in the left premaxilla (Fig. 3A-B). In the mandible a total of five incisors are preserved, with two on the right and three on the left (Fig. 3C-D). The tooth crowns of all the upper incisor are broken at the level of the alveolus except for I2, that is broken just after the level of the alveolus. The lower incisors have received most of the damage with all the teeth showing breakage along the shaft or being broken at alveolus level (Fig. 3C-D). There is no replacement activity occurring in either the lower or upper incisors. Although there is no evidence of remnant roots in the premaxillae, it is likely that the preserved upper incisors had replaced a previous tooth generation (Fig. 3A-B). All upper incisors have open roots, which suggests that they were still developing. The Incisors are showing signs of alternating tooth replacement activity.

Canines- there are two functional upper canines, one the left and one on the right. The upper canines are well preserved although the left canine is broken around the neck, the tooth crowns are preserved (Fig. 3A-B). The alveolus of both canines is small showing that the teeth were fully erupted. Only the right upper canine show signs of tooth replacement, with the remnant root of the

previous canine preserved distal to the functional canine (Figs 2 and 3B), there is no replacement tooth seen in this specimen. The roots of both upper canines are closed indicating that the teeth are fully developed. In the mandible there are two canines preserved; the left canine is well preserved whereas the right one is broken at the level of the margin of the alveolus. There is no replacement activity associated with the lower canines, the alveolus is small, and the roots are closed indicating that the tooth is fully developed (Fig. 3C-D). The canines are showing signs of alternating tooth replacement activity.

Figure 2: Virtual horizontal section through the anterior dentition of BP/1/155 an indeterminate gorgonopsian. Abbreviations: were C1 and C2, is the functional upper canine; c1-2 is the lower canine; RR, remnant root of an old canine, I1-5 and i1-5 is the upper and lower incisors.

Postcanines- The postcanines are not well preserved. In the right maxilla, seven postcanines are preserved (Fig. 3C) and in the left one, only one postcanine is preserved (Fig. 3D). The tip of PC6 is broken mid crown. There is no replacement activity associated with maxillary postcanines on either side, with the right PC2, PC4, PC6, PC7, and left PC1 having closed roots (Fig. 3A-B). On the mandible three postcanines are preserved on the right and six post canines are preserved on the left (Fig. 3C-D). There is no replacement activity associated with the mandibular postcanine teeth, with all the lower postcanines exhibiting closed roots. The postcanines are showing signs of alternating tooth replacement activity.

Figure 3: Three-dimensional rendering of the dentition of specimen BP/1/155 an indeterminate gorgonopsian, A upper right; B, upper left; C, lower right; D, lower left. Abbreviations: I 1-5 present the upper incisor of both the left and the right maxillary teeth, i1-4 represent the lower incisors of both the left and right mandibulae teeth, C1-2 & c1-2 represent the upper and lower canines, RR is a remnant root of a previous tooth that is in the process of being resorbed, PC 1-7 represent the maxillary post canines of both the left and right side and lastly pc1-6 represent the mandibulae post canines of both the left and right side.

Specimens with type 1 replacement

AM 3752 (Eriphostroma)

Description:

Incisors: Looking at the transverse slice, on the right premaxilla only four incisors are present, and only one on the left side. The upper incisors have no replacement activity. On the mandible, four incisors are preserved on the left and two are preserved on the right. The lower incisors show no signs of replacement activity. The incisors are showing alternating tooth replacement

Canine: There one functional upper canine on each side of the premaxilla, both show signs of tooth replacement, with each canine having a remnant root located distally from the functional canine. From the transverse slice the lower canines (c1 and c2) show no signs of replacement. The specimen is showing pattern of alternating tooth replacement. The canines are indicating alternative tooth replacement.

Postcanines: The upper and lower postcanines are poorly preserved, with no replacement activity. The post canines are considerably smaller as compared to the incisors and canines

Figure 4: Virtual horizontal section through the anterior dentition of AM 3752 a Eriphostroma. A is the upper jaw of the specimen. A is the upper jaw and B is the lower jaw, Abbreviations were C1 and C2, is the functional upper canine; c1-2 is the lower canine; RR, remnant root of an old canine, I1-5 and i1-5 is the upper and lower incisors, PC1-2 and pc1-3 is the lower and upper postcanines.

BP/1/216 (Arctops kitchingi)

Description:

Incisors: On the maxilla the specimen has a total of 10 maxillary incisor teeth, five on both sides. Only the left premaxillary incisors have replacements. I3 has a remnant root of the previous incisor and I5 has a replacement tooth located distal lingually of the functional incisor, the placement tooth is significantly smaller. The incisors are indicating alternative tooth replacement.

Canines: There is one functional upper canine on each side both show signs of tooth replacement, with each canine having a remnant root located distally from the functional canine. The specimen is showing pattern of alternating tooth replacement. The canines are indicating alternative tooth replacement.

Postcanines: Only 3 upper left postcanines and 4 upper right postcanines are preserved, with no replacement activity. The post canines are considerably smaller as compared to the incisors and canines.

Figure 5: Virtual horizontal section through the anterior dentition of BP/1/216 an Arctops kitchingi. A is the upper incisors and canines, B is the upper postcanines. Abbreviations: were C1 and C2, is the functional upper canine; c1-2 is the lower canine; RR, remnant root of an old canine, I1-5 and i1-5 is the upper and lower incisors.

Type 2. One functional canine (distal), one replacement (mesial)

BP/1/673 (Cynarioides gracilis)

Description: The specimen is a complete skull with the mandible preserved in tight occlusion (Fig. 7), most the teeth are preserved in the jaw. The skull shape is mostly preserved in its original shape with only a slight compressing on the occipital region of the skull.

Incisors- There are 10 incisors preserved in the premaxilla, with five on each side. The upper incisors are preserved in good condition, with the tooth crown preserved in all the upper incisors. There is a lot of replacement activity associated with the upper incisors, with I1, I2, I4 on the right and on the left I1, I2, I4 all have replacement teeth located lingually (Fig. 7A-B). There is a clear shape change of the neck of the tooth in all the functional incisors that are associated with replacements. The shape has changed from a smooth rounded surfaces to a more oval – bean shape surface e.g. I4 (Fig. 6). All the upper incisors having open roots, I4, I5 on the right and on the left I5 showing more evidence of root resorption on the roots (Fig. 7A-B). Replacement activity is also seen in the upper incisor in the form of a large alveolae seen in incisors on the left I1, I2, I4 and right I3 and I4 as compared to I5 (Fig. 6). Another indicator to tooth replacement activity seen in the upper incisor is the thickness of the enamel. I1 and I3 on the left, I1 and I15 on the right are much thinner as compared to I5 (Fig. 6). The incisors are showing pattern of alternating tooth replacement.

There is a total of six mandibular incisors preserved, with only two incisors on the right (i1 and i2; Fig. 7C) and 4 preserved in the left (i1–i4; Fig. 7D). Only the left mandibular incisors are preserved in good condition the tooth crown preserved. In the right mandibular teeth i1 is broken on the neck and i2 the tooth crown is not preserved (Fig. 7C-D). There is very little replacement activity in the lower incisors, with only i2 and i4 having replacement teeth associated with them (Fig. 7C-D). Only i1, i2, i4 have open roots suggesting the teeth are not fully developed, i3, i5, i6 all have closed roots. The upper and lower incisors are showing pattern of alternating tooth replacement.

Canines- there one functional canine on both side of the maxilla. The upper canines are preserved in good condition, with the tooth crown preserved (Fig. 7A-B). Both the right and left upper canine show signs of tooth replacement. There is remnant root of the previous canine preserved mesially on the left functional canine, and lingually on the right functional teeth (Figs 6 and 7A-B). Both canines are associated with replacement teeth, located mesial to the functional tooth on the right (RC1) and lingual to the functional tooth on the left (RC2). The upper canines have open root that indicate that the teeth are not fully developed. In the mandible the right canine is well preserved, whereas the left is broken along the neck of the canine. Both lower canines show evidence of replacement activity. The lower canines having replacement teeth associated with them. The roots are open indicating that the tooth is not fully developed (Fig. 6C-D). The canines are showing pattern of alternative tooth replacement.

FIGURE 6: Virtual horizontal section through the anterior dentition of BP/1/673 a Cynarioides gracilis. Abbreviations: were C1 and C2, is the functional upper canine; c1-2 is the lower canine; RR, remnant root of an old canine, I1-5 and i1-5 is the upper and lower incisors. RC1-2 represent replacement teeth

Postcanines- The postcanines are well preserved. The specimen has 14 preserved postcanines, nine on the right and six on the left. There is a tooth missing between PC4 and PC6, which may have fallen out. Replacement activity is only seen in the left maxillary teeth, with the left PC3 having a replacement tooth. All the upper postcanines have open roots, suggesting the postcanines are not fully developed except for PC6, PC9 and on the left PC2, and PC4 look to have closed (or very nearly closed) roots. Only the left mandibular postcanines are preserved in good condition, with the right mandibular postcanine teeth broken at alveolus level and along the neck of the tooth. Replacement activity is only seen in the left and right pc6, having a replacement tooth associated with both postcanines. All the lower postcanines have open roots, suggesting the postcanines are not fully developed. The postcanines are showing pattern of alternative tooth replacement.

Figure 7: Three-dimensional rendering of the dentition of specimen BP/1/673 a Cynarioides gracilis, A upper right; B, upper left; C, lower right; D, lower left. Abbreviations: I 1-5 present the upper incisor of both the left and the right maxillary teeth, i1-6 represent the lower incisors of both the left and right mandibulae teeth, C1-2 & c1-2 represent the upper and lower canines, RR is a remnant root of a previous tooth that is in the process of being resorbed, RC1-2 and rc 1-2 represent replacement teeth, PC 1-8 represent the maxillary post canines of both the left and right side and lastly pc1-9 represent the mandibulae post canines of both the left and right side.

Type 3. One functional canine (distal), multiple replacements (mesial & distal)

BP/1/709 (Aloposaurus)

Description: The specimen is a complete skull with the mandible preserved in tight occlusion (Fig. 9), most the teeth are preserved in the jaw, the skull is compressed on the right temporal region of the skull, this has led to misplacement of teeth that made it difficult to segment the upper right teeth.

Incisors- There are a total of four preserved teeth in the premaxilla. The upper incisors are preserved in good condition, with the tooth crown preserved in all the upper incisors. There is a lot of replacement activity associated with the upper incisors, with I1, I2, I3, I4 having replacement teeth located lingually (Fig. 9A). All the upper incisors having open roots, indicating that the teeth are not fully developed. There is a total of 8 mandibular incisors preserved, with four on each side (Fig. 9C- D). None of the mandibular incisors are well preserved, with most being broken at the tooth crown (Fig. 9C-D). Replacement activity is seen in the lower incisors, with only i2 and i4 on the right and i2 and i4 on the left having replacement teeth associated with them (Fig. 9C-D). All the mandibular incisors have open roots showing that the teeth are not fully developed. The upper and lower incisors are showing pattern of alternating tooth replacement.

Canine- There are two upper canines. The upper canines are preserved in good condition with the tooth crown preserved (Fig 8 A). The two upper canines show signs of tooth replacement, with both canines having replacements teeth positioned dorsal lingual to the functional canine (Figs 8 and 9A). The canines have open roots, indicating that the teeth are not fully developed. In the mandible the right canine is well preserved, whereas the left is broken along the neck. Both lower canines show evidence of replacement activity, with both canines having replacement teeth associated with them. The roots are open indicating that the teeth were not fully developed (Fig. 9B-C). The upper and lower canines are showing pattern of alternating tooth replacement.

Figure 8: Virtual horizontal section through the anterior dentition of BP/1/709 a Aloposaurus. Abbreviations: were C1.1 and C1.2, is the functional upper canine.c1-2 is the lower canines, I1-4 and i1-4 is the upper and lower incisors; RC1-2 represent replacement teeth.

Postcanines- Only two of the upper left postcanines are preserved in good condition, and no replacement activity is associated with the teeth (Fig. 9A). All the upper postcanines have open roots, suggesting the postcanines are not fully developed. A total of nine mandibular postcanines are preserved in good condition, with three on the right and six on the left mandible (Fig. 9B-C). No replacement activity is seen on either side of the mandibular postcanine series. All the lower postcanines have open roots, suggesting the postcanines are not fully developed. The postcanines are showing pattern of alternating tooth replacement.

Figure 9: Three-dimensional rendering of the dentition of specimen BP/1/709 an Aloposaurus, A upper left; B, lower right; D, lower left. Abbreviations: I 1-5 present the upper incisor of both the left

and the right maxillary teeth, i1-4represent the lower incisors of both the left and right mandibulae teeth, C1-2 & c1-2 represent the upper and lower canines, RC1-2 and rc 1-2 represent replacement teeth,PC 1-2 represent the maxillary post canines of both the left and right side and lastly pc1-6 represent the mandibulae post canines of both the left and right side.

Specimen with type 3 replacement

Specimen BP/1/575 (Aelurosaurus)

Description:

Incisors: The specimen is incomplete skull with the mandible not preserved in (Fig. 10). Looking at the transverse slice, on the right premaxilla only five incisors are present, and only four on the left side. Replacement activity is only seen in I4 in the upper incisors. The upper and lower incisors are showing pattern of alternating tooth replacement.

Canine: There are two upper canines. The two upper canines show signs of tooth replacement, with both canines having replacements teeth positioned dorsal lingual of each canine and lingually on the maxilla. The canines are showing pattern of alternating tooth replacement.

Postcanines: none of the postcanines are preserved

Figure 10: Virtual horizontal section through the anterior dentition of BP/1/709 a Aelurosaurus Abbreviations: were C1.1 and C1.2, is the functional upper canine.c1-2 is the lower canines, I1-4 and i1-4 is the upper and lower incisors; RC1-2 represent replacement teeth.

Discussion

Variation in dental replacement during ontogeny in Gorgonopsians

Dental replacement patterns in gorgonopsians vary across ontogenetic stages, with juveniles showing higher rates of tooth replacement compared to adults. This trend is evident in specimens such as BP/1/673 (Cynarioides gracilis) and BP/1/709 (Aloposaurus), both which are identified as juveniles. In these specimens, open-rooted canines and incisors suggest continuous development (Sassoon et al., 2015; Norton, 2021). Having replacement teeth positioned mesially and distally relative to the functional teeth (Fig.6-9). The presence of multiple replacement teeth supports a high turnover rate (Haridy et al.,2018), likely necessary to accommodate rapid jaw growth and sustain sharp teeth for prey capture. In the juvenile specimen BP/1/673, replacement incisors I1, I2, and I4 are positioned lingually to the functional teeth, while the canines show replacement activity mesially, showing a dynamic cycle of replacement during early growth stages (Fig. 6). Similar replacement patterns have been seen in early cynodonts and therocephalians, in which high replacement rates in juveniles help maintain an effective dentition as jaws expand with age (Benoit et al., 2017; Norton et al., 2021).

In contrast, sub-adult and adult specimens, including BP/1/155 (an indeterminate gorgonopsian) and BP/1/575 (Aelurosaurus), show reduced replacement activity. Both specimens display closed-rooted canines and postcanines, suggesting that replacement cycles had slowed or stopped altogether by this stage (Abdala et al., 2023). For instance, in BP/1/155, the presence of a distal remnant root without an accompanying replacement tooth indicates a shift toward a more stable dentition (Fig. 2-3). This pattern aligns with observations in adult cynodonts, where replacement decreases significantly, likely reflecting a shift from the high metabolic demands of early growth to a more energy-efficient system in adulthood (Hopson & Kitching, 2001). This ontogenetic shift is observed in other therapsid taxa as well, where juvenile specimens show high replacement rates that slowdown in adulthood, resulting in a more durable, stable dentition suited for predatory roles (Abdala et al., 2013; Van den Brandt & Abdala, 2018).

The variation in replacement rates between juvenile and adult gorgonopsians aligns with the evolutionary pressures facing therapsid predators. Juveniles likely required frequent tooth replacement to support rapid growth and to sustain sharp, functional teeth, as observed in juvenile gorgonopsians like BP/1/673 and BP/1/709. In contrast, the reduced replacement in adults, such as BP/1/575 may reflect a shift toward maintaining durable teeth capable of withstanding greater wear and tear over time without the need for frequent replacement a pattern. This is also seen in derived

non-mammalian synapsids and contributing to the transition toward mammalian diphyodonty (Kermack & Medawar, 1997; Hopson, 2011).These findings are consistent with the notion that tooth replacement patterns may have evolved in response to developmental needs and dietary pressures, with gorgonopsians displaying an early ontogenetic stage characterized by high replacement activity, which diminishes as individuals approach adulthood and their dentition stabilizes for predatory efficiency.

The double canine condition in Gorgonopsians

The hypothesis that primitive gorgonopsians retained a double canine condition was tested by comparing juvenile and sub-adult specimens as well as examining differences between basal and derived taxa. The data reveal that the double canine condition is most pronounced in juvenile specimens, such as BP/1/709 (Aloposaurus). Where multiple replacement teeth are observed mesial and distal to the functional canine, suggesting a temporary “double canine” state during early ontogeny (Fig. 9). In BP/1/709, both upper canines exhibit replacement activity from two positions, indicating the presence of additional canines and replacements throughout juvenile development. This pattern may reflect the high metabolic demands of rapid growth stages, as continuous replacement could help maintain sharp, functional teeth essential for prey capture and defence (Norton et al., 2021; Hopson & Kitching, 2001).

In contrast, adults such as BP/1/575 (Aelurosaurus) display only a single functional canine, with closed roots and no active replacement. Suggesting that the double canine condition stops as gorgonopsians mature (Fig. 10). The closed-rooted canines in BP/1/575 and the absence of distal replacements suggest that the juvenile double canine condition is a developmental trait that becomes simplified in adulthood (Haridy et al., 2018). The loss of double canines in mature individuals may reflect a shift toward energy efficiency, where the metabolic demands of continuous replacement cycles are minimized as the animal stabilizes its dentition. Similar patterns are observed in other therapsids, where early life stages show a higher incidence of replacement, especially in canines, that later stabilizes to support a more energetically sustainable adult morphology (Abdala et al., 2013; Van den Brandt & Abdala, 2018).

When examined across a phylogenetic framework, the double canine condition is more characteristic of certain derived gorgonopsians from the Late Permian rather than basal gorgonopsians. Specifically, double canines are observed in BP/1/709 (Aloposaurus) and BP/1/575 (Aelurosaurus), both of which are derived taxa with active replacement in the upper dentition. This contrasts with the simpler, single-canine arrangement seen in basal Middle Permian gorgonopsians, where replacement cycles are more limited. For example, in the more basal Cynarioides gracilis (BP/1/673) and the most basal Eriphostoma (AM 3752), only have a single mesial replacement is associated with the functional canine, indicating a simpler dental structure without the double canine condition (Fig. 7A-B).

These findings differ from patterns observed in other early therapsids, where frequent replacement cycles and the presence of multiple canines are typically seen in basal taxa to support dietary demands during rapid growth (Hopson & Kitching, 2001). However, the appearance of double canines in derived gorgonopsians suggests that this condition may have evolved independently as an adaptation to specific ecological pressures within this lineage. Derived taxa, such as Aloposaurus and Aelurosaurus, may have retained this feature to maintain functional dentition despite high wear associated with a predatory lifestyle in later Permian ecosystems (Norton et al., 2021). This trend contrasts with findings in early cynodonts and therocephalians, where reduced canine replacement and a single functional canine reflect an evolutionary move toward more mammal-like dentition (Kermack & Medawar, 1997; Benoit et al., 2017; Hopson, 2011; Norton et al., 2021).Overall, the presence of double canines in derived gorgonopsians rather than basal members highlights a unique evolutionary path within this group, suggesting that the condition may have re-emerged in response to specific functional demands in the late Permian environments.

Double canines are found in several non-gorgonopsian synapsid groups, especially among early therapsids and primitive cynodonts, and are typically associated with high replacement rates in juvenile stages and continuous replacement cycles. Juvenile therocephalians, such asLycosuchus, display a double canine condition with multiple replacement teeth at different developmental stages, a pattern that reduces as individuals mature a trend also observed in gorgonopsians (Abdala et al.,

2013; Hopson, 2011). This ontogenetic shift in both therocephalians and gorgonopsians indicates that high replacement rates and double canines were primitive traits in therapsids, likely needed to accommodate growth and dietary demands in young animals. Similarly, early cynodonts, such as Thrinaxodon and Galesaurus, exhibit double canines in juveniles, but as they mature, their dentition stabilizes to only one functional canine (Norton et al., 2021; Cabreira, et al., 2022).

The juvenile presence of double canines in both gorgonopsians and cynodonts may suggests that frequent replacement was important for these early carnivores. This trends towards the mammalian condition of a single canine per tooth locus, implying an adaptation across therapsids toward more energy efficient dentition (Kermack & Medawar, 1997; Hopson, 2011). Additionally, some basal biarmosuchians, such as Hipposaurus, also retain double canines in juveniles, but these stabilize to a single canine in adults, similar to patterns in therocephalians and cynodonts (Hopson & Kitching, 2001). This pattern in biarmosuchians suggests that double canines were likely an ancestral synapsid feature, conserved in multiple lineages and gradually reduced as specialized dentition evolved.

The consistent reduction of double canines in adults aligns with findings from studies on sabre-toothed cats, where the functional significance of maintaining a single pair of large canines was explored using biomechanical models. One such study by McHenry et al. (2007) investigated the mechanical performance of the canines in Smilodon fatalis, demonstrating that the size and morphology of these teeth were optimized for delivering precise, high-force bites rather than enduring repeated stress. The study suggested that maintaining a single pair of robust canines allowed for efficient energy allocation and minimized the risk of structural failure during predation, supporting their specialized feeding strategy (McHenry et al.,2007).

Applying this perspective to gorgonopsians, the reduction of double canines in adults may reflect a similar evolutionary pressure to enhance the mechanical efficiency of the functional canines. Derived gorgonopsians, such as Aloposaurus and Aelurosaurus, which retain double canines in juveniles, likely transitioned to a single functional canine in adults to optimize bite strength and stability. While gorgonopsian canines are not as prominent as those of sabre-toothed cats, the biomechanical principles of stress distribution and energy conservation may similarly explain the evolutionary trend toward single functional canines in adults. Future research incorporating biomechanical modelling in gorgonopsians could further explain the role of functional demands in shaping their dental evolution. The juvenile double canines observed in gorgonopsians, therocephalians, cynodonts, and biarmosuchians suggest that high replacement rates were ancestral features within therapsids, while the eventual reduction of double canines in adults signals an adaptive shift toward stable, low-maintenance dentition. This trend is seen in the mammalian condition, where a single set of canines is the normal (Van den Brandt & Abdala, 2018).

Alternating dental replacement in Gorgonopsians: ontogenetic and phylogenetic differences

Although limited by the sampling, alternating dental replacement, where adjacent teeth are replaced in an offset pattern, is evident in gorgonopsians and seem to vary according to ontogenetic stage. In juvenile specimens, such as BP/1/673 (Cynarioides gracilis) and BP/1/709 (Aloposaurus), alternating replacement is prominent in both the incisors and postcanines, as indicated by the presence of replacement teeth adjacent to functional teeth along the dental rows (Fig. 6-9). This staggered replacement minimizes the disruption of functional dentition during rapid growth, which may have been essential for juveniles requiring constant, sharp teeth for effective feeding (Cisneros et al., 2011). This pattern aligns with findings in other basal therapsids, where alternating replacement in juveniles supports feeding efficiency and dental functionality as jaws expand (Abdala et al., 2013). Alternating replacement has also been documented in non-mammalian synapsids, such as therocephalians and early cynodonts, where replacement ensures that functional teeth are consistently maintained throughout juvenile development (Norton et al., 2021; Hopson & Kitching, 2001).

In contrast, more derived gorgonopsians, particularly the Late Permian rubidgeines, show a reduced incidence of alternating replacement (Cisneros et al., 2011 and Cabreira, et al., 2022). For example, AM 3752 (Eriphostroma), a basal Middle Permian gorgonopsian, exhibits pronounced alternating replacement, particularly in the incisors. Which is consistent with the primitive therapsid

condition. In contrast, the rubidgeines display a more stable dentition, with a less frequent replacement cycle and fewer signs of alternating replacement, particularly in the postcanines and canines (Abdala et al., 2023; Norton, 2021 and McCollum et al., 2001). This shift suggests a movement toward a simplified replacement pattern, likely driven by selective pressures that favoured reduced tooth turnover and enhanced dental stability in mature individuals (Kermack & Medawar, 1997). In the Late Permian BP/1/155 specimen, the lack of pronounced alternating replacement, prominent in the postcanines, indicates that tooth replacement had slowed as dentition became more specialized, reflecting a trend towards energy conservation and durability of teeth in adults.

The evolutionary reduction in alternating replacement in gorgonopsians is similar to patterns observed in derived therapsids and early mammals, where stabilized dentition minimizes metabolic costs associated with frequent replacements (McCollum et al., 2001). In derived cynodonts, for example, alternating replacement becomes less common, with many species showing signs of a shift towards a single replacement cycle or diphyodonty, similar to what is eventually seen in mammals (Hopson, 2011; Cabreira, et al., 2022). This evolutionary trend from frequent, alternating replacement in basal taxa to a more stabilized dentition in derived taxa may indicate a general trend towards dental stability in mature stages. For gorgonopsians, this pattern suggests that alternating replacement was an ancestral condition, that may have been gradually reduced in more derived lineages as they adapted to the ecological demands of predation with a more stable, functionally consistent dentition (Van den Brandt & Abdala, 2018). The alternating dental replacement observed in juvenile gorgonopsians such as BP/1/673 and BP/1/709 highlights a primitive feature essential for maintaining functional dentition during growth, while the reduced frequency of alternating replacement in derived taxa aligns with broader therapsid trends favouring stabilized dentition for energy efficiency (O'Meara et al., 2018).

Conclusion

This study examined dental replacement patterns in six gorgonopsian specimens from the Middle to Late Permian to investigate the evolutionary implications of tooth replacement variation across ontogenetic and phylogenetic stages. This study supports ontogenetic variation in tooth replacement in gorgonopsians. Juveniles such as BP/1/709 (Aloposaurus) and BP/1/673 (Cynarioides gracilis) exhibit higher replacement rates and prominent alternating replacement in the incisors and postcanines, likely reflecting the need for continuous functional dentition during rapid growth. The double canine condition, unexpectedly, is more pronounced in derived taxa rather than in basal Middle Permian gorgonopsians. This double canine trait in derived specimens, such as Aloposaurus and Aelurosaurus, contrasts with other therapsid groups that show a trend towards a single, canines with reduced replacement rates.

The observed ontogenetic and phylogenetic differences in gorgonopsian dental replacement patterns suggest that these therapsids retained primitive traits, such as high juvenile replacement rates, while also developing unique adaptations like the double canine condition in later forms. This duality underscores the complexity of therapsid dental evolution, where gorgonopsians diverged from other lineages by retaining high replacement activity in derived members. These findings provide insight into the adaptive flexibility of gorgonopsians, highlighting a transitional stage between reptilian and mammalian dental characteristics and contributing to our understanding of the evolutionary paths that led to the specialized dentition seen in mammals.

This study's limited to a small sample size of six gorgonopsian specimens, restricting the generalization of findings across the clade and highlighting the need for a broader sample from diverse geological contexts. Limited specimens across ontogenetic stages also constrain insights into developmental changes in dental replacement patterns. Although comparisons were made between basal and derived gorgonopsians, a more comprehensive phylogenetic context that includes additional non-gorgonopsian therapsids would better clarify whether observed patterns are unique to gorgonopsians or part of a broader evolutionary trend. The evolutionary purpose of the double canine condition remains unknown, as its functional significance is uncertain without further data on dietary habits or functional morphology. Similarly, while alternating replacement patterns were observed, the mechanisms driving this pattern whether for feeding efficiency or growth require further study in a broader therapsid sample. Fossil preservation issues, including postmortem

damage and partial fossilization, introduce potential biases that may obscure detailed dental features. The lack mandibles for some fossils limits a comprehensive analysis of full jaw replacement cycles, potentially not showing gorgonopsian complete dental complexity.

References